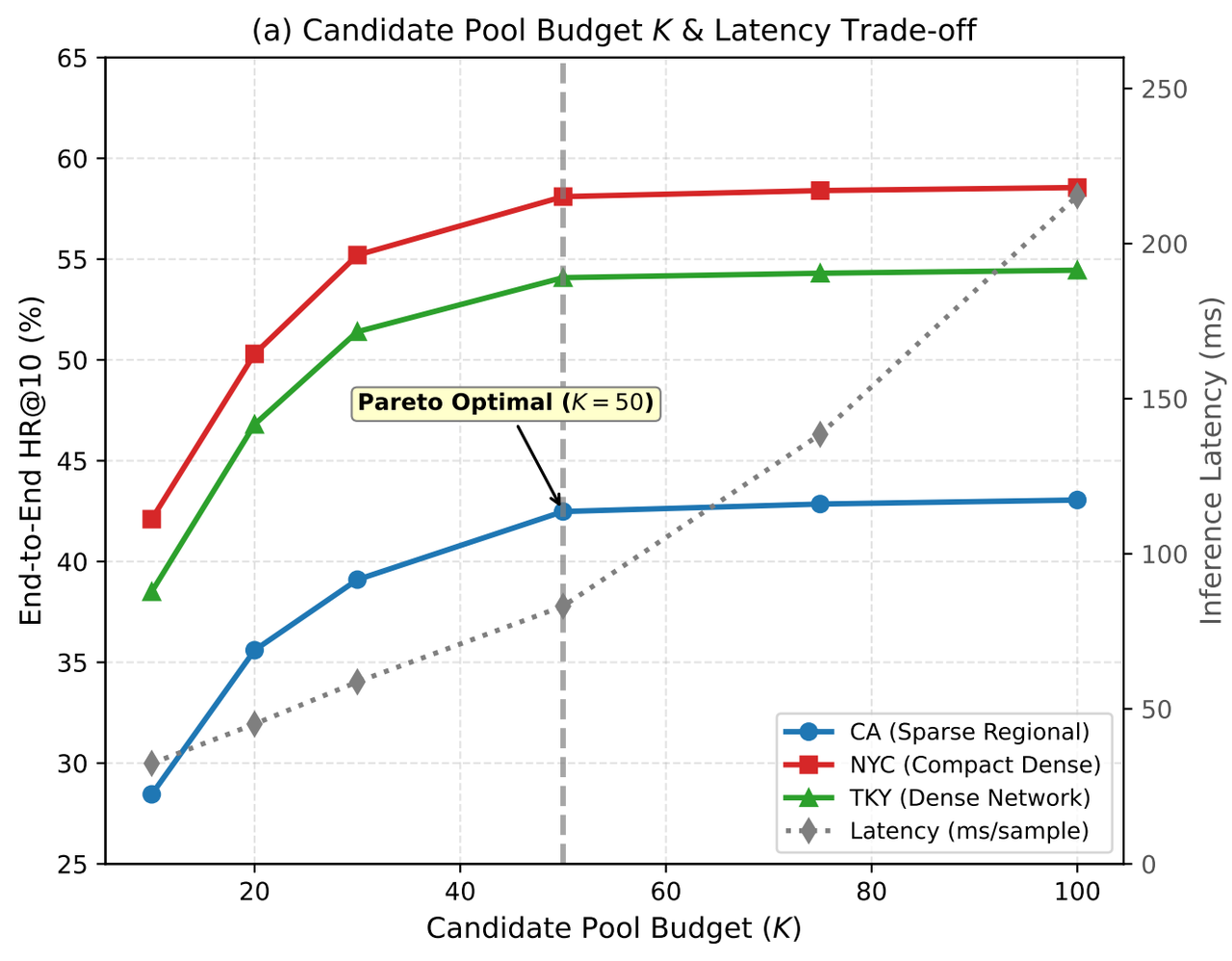
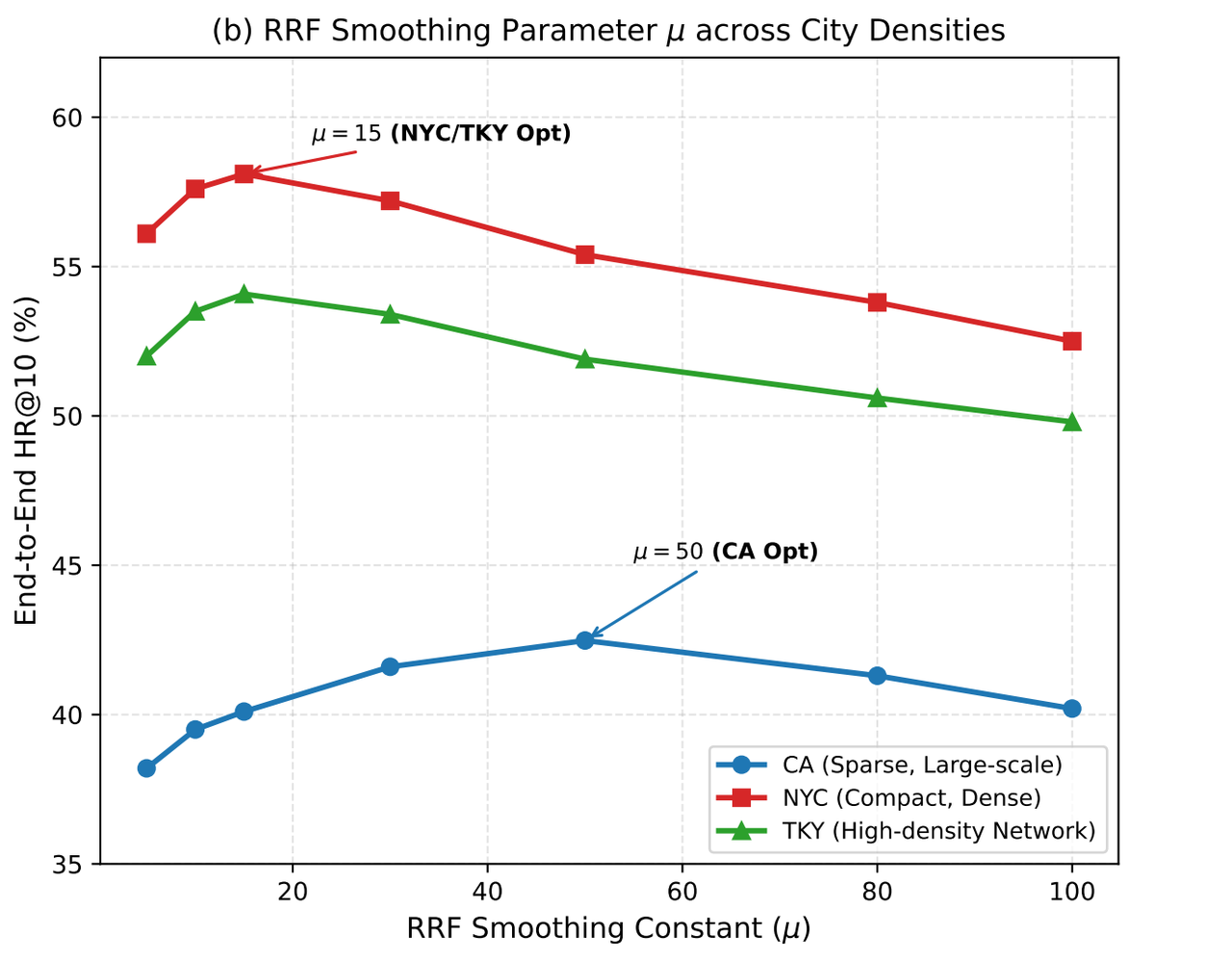
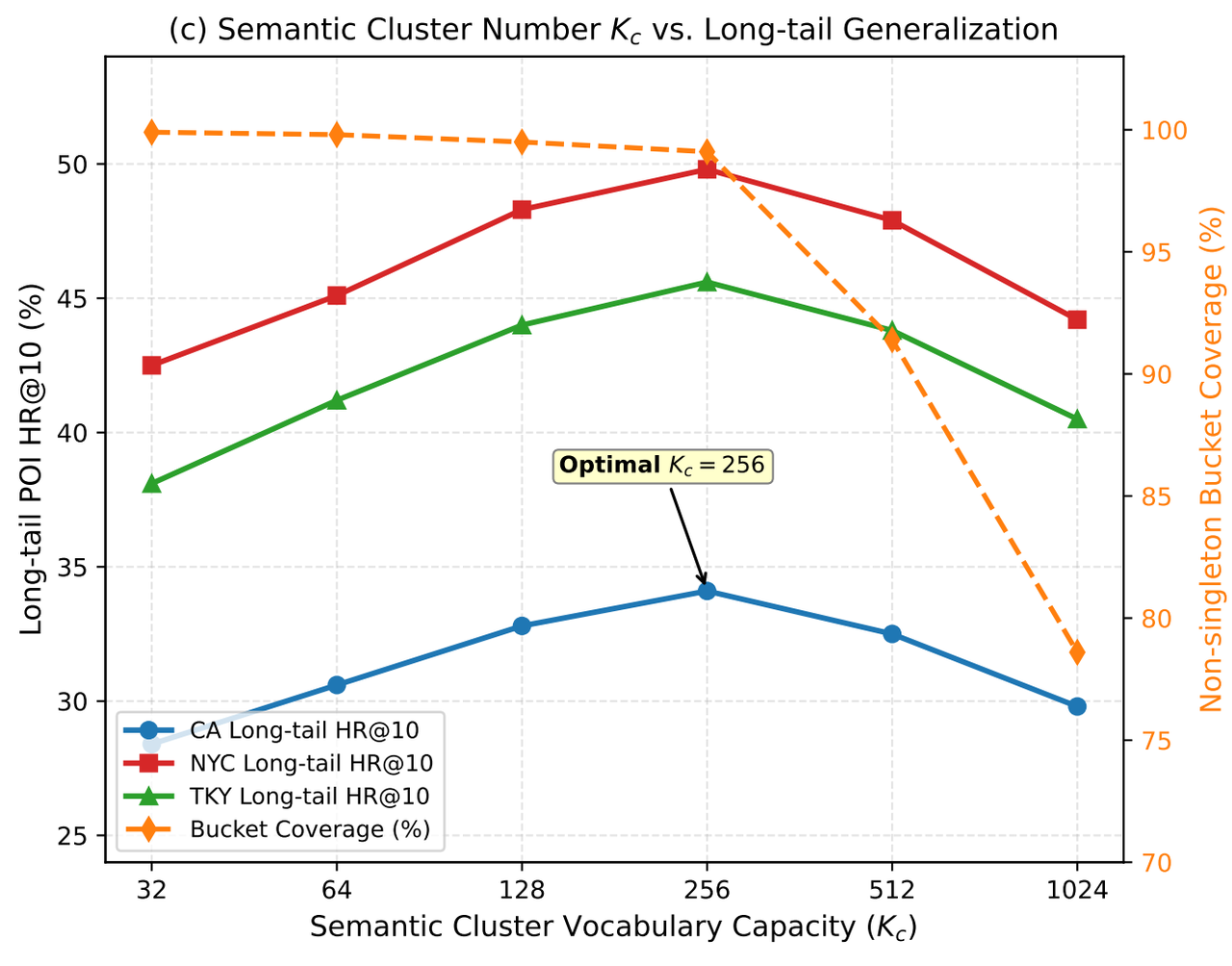
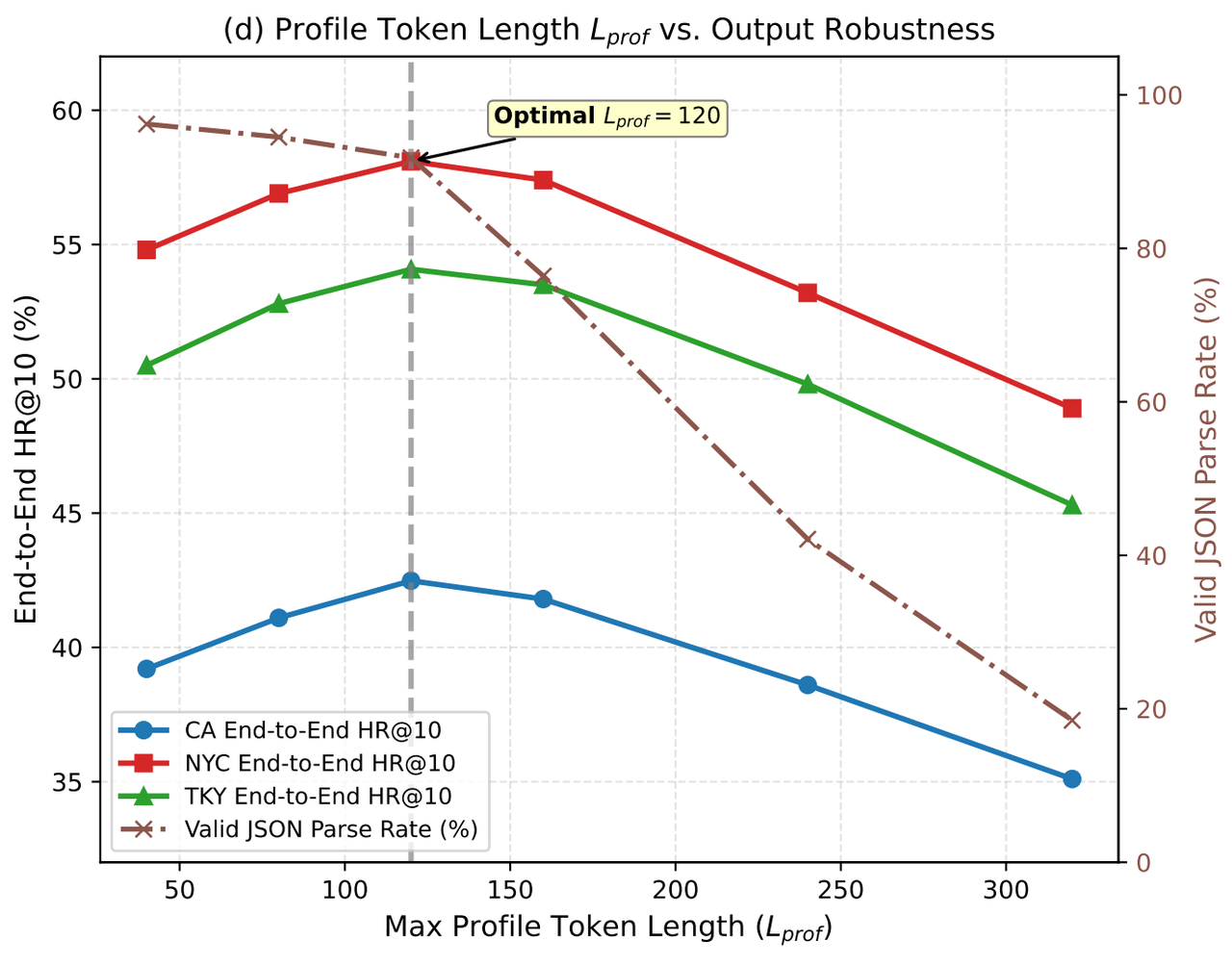

(a) Candidate Pool Budget K & Latency Trade-off
End-to-End HR@10 (%)
Inference Latency (ms)
Candidate Pool Budget (K)
Pareto Optimal ($K = 50$)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)


(b) RRF Smoothing Parameter μ across City Densities
End-to-End HR@10 (%)
RRF Smoothing Constant (μ)
$\mu = 15$ (NYC/TKY Opt)
$\mu = 50$ (CA Opt)
CA (Sparse, Large-scale)
NYC (Compact, Dense)
TKY (High-density Network)


(c) Semantic Cluster Number K_c vs. Long-tail Generalization
Long-tail POI HR@10 (%)
Non-singleton Bucket Coverage (%)
Semantic Cluster Vocabulary Capacity (K_c)
Optimal $K_c = 256$
CA Long-tail HR@10
NYC Long-tail HR@10
TKY Long-tail HR@10
Bucket Coverage (%)


(d) Profile Token Length L_{prof} vs. Output Robustness
End-to-End HR@10 (%)
Valid JSON Parse Rate (%)
Max Profile Token Length (L_{prof})
Optimal $L_{prof} = 120$
CA End-to-End HR@10
NYC End-to-End HR@10
TKY End-to-End HR@10
Valid JSON Parse Rate (%)